

GLOBAL SUPPLY CHAIN & LOGISTICS PERFORMANCE

Final Project Report

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Chapter 1

Introduction

1.1 Project Overview

The Global Supply Chain & Logistics Performance project focuses on analyzing logistics and supply chain operations using the DataCo Supply Chain Dataset. The project applies data analytics techniques to identify business insights, operational trends, customer behavior patterns, and shipping performance metrics.

1.2 Background of the Study

Modern organizations depend heavily on efficient supply chain systems for inventory management, transportation planning, and customer satisfaction. Data analytics plays a significant role in optimizing supply chain performance and reducing operational costs.

1.3 Problem Statement

Organizations often face challenges such as shipment delays, inventory imbalances, inefficient transportation planning, and fluctuating customer demand. These challenges impact overall business performance and profitability.

1.4 Project Objectives

- Understand the structure of the supply chain dataset.
- Analyze logistics and shipping operations.
- Identify sales and profit trends.
- Examine delivery performance.

- Evaluate customer behavior.
- Generate business insights for decision making.

1.5 Scope of the Project

The project focuses on supply chain analytics using historical order and shipment data. It includes data preprocessing, visualization, trend analysis, and insight generation.

1.6 Project Deliverables

- Cleaned Dataset
- Data Analysis Report
- Visualizations
- Business Insights
- Final Project Documentation

1.7 Report Organization

This report is organized into five chapters covering project introduction, requirements analysis, data understanding, preprocessing, and analytics.

Chapter 2

Requirement Analysis

2.1 Domain Study

Supply Chain Management involves planning, procurement, inventory control, warehousing, transportation, and customer service activities.

2.2 Existing System Analysis

Traditional systems rely heavily on manual reporting and monitoring, making it difficult to identify inefficiencies and predict future trends.

2.3 Proposed System

The proposed system uses Python-based analytics to study logistics data and generate meaningful business insights through data visualization.

2.4 Functional Requirements

- Dataset Upload
- Data Cleaning
- Data Analysis
- Visualization Generation
- Report Creation

2.5 Non-Functional Requirements

- Accuracy
- Reliability
- Scalability
- Maintainability
- Performance

2.6 Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab
- GitHub

Chapter 3

Data Collection and Understanding

3.1 Data Source

The DataCo Supply Chain Dataset was used for this project.

3.2 Data Description

The dataset contains customer information, product details, sales data, shipping information, delivery status, markets, and regions.

3.3 Data Structure

The dataset consists of rows representing orders and columns representing attributes related to customers, products, and logistics.

3.4 Data Dictionary

Attribute	Description
Order ID	Unique Order Identifier
Customer ID	Customer Identifier
Product Name	Product Purchased
Sales	Sales Amount
Profit	Profit Generated
Market	Market Region
Shipping Mode	Delivery Method
Delivery Status	Shipment Status
Order Date	Date of Order

Country	Customer Country
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3.5 Data Quality Assessment

The dataset was assessed for missing values, duplicate records, inconsistent entries, and outliers before analysis.

Chapter 4

Data Preprocessing

4.1 Data Cleaning

Data cleaning was performed to remove inconsistencies and improve data quality.

4.2 Handling Missing Values

Missing values were identified and appropriately handled using suitable techniques.

4.3 Duplicate Removal

Duplicate records were removed to avoid biased analysis.

4.4 Data Transformation

Columns were transformed into appropriate formats for analysis.

4.5 Data Validation

The cleaned dataset was validated to ensure accuracy and consistency.

4.6 Final Prepared Dataset

The final dataset was prepared for exploratory data analysis and visualization.

Chapter 5

Data Analysis and Modeling

5.1 Exploratory Data Analysis (EDA)

5.1.1 Delivery Status Distribution

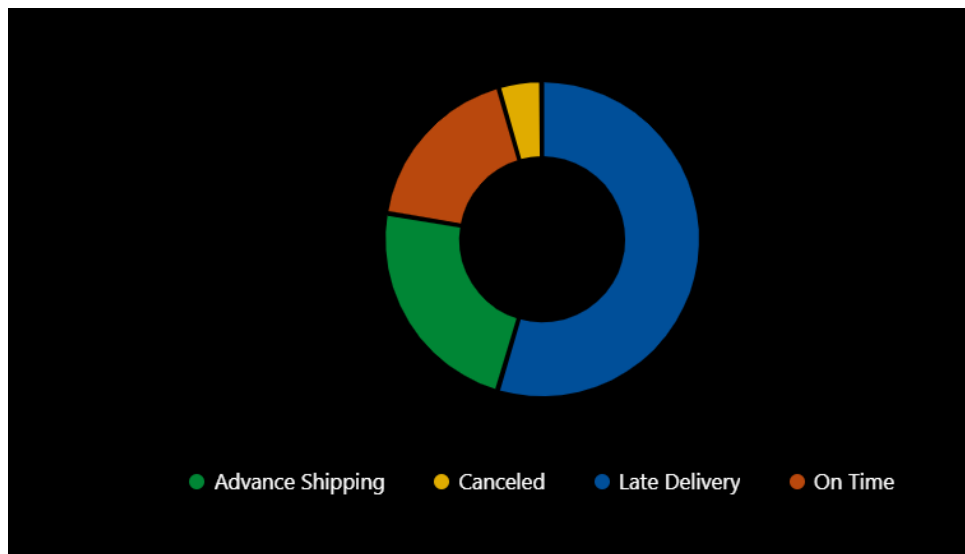


Figure 5.1: Delivery Status Distribution

5.1.2 Customer Segment Distribution

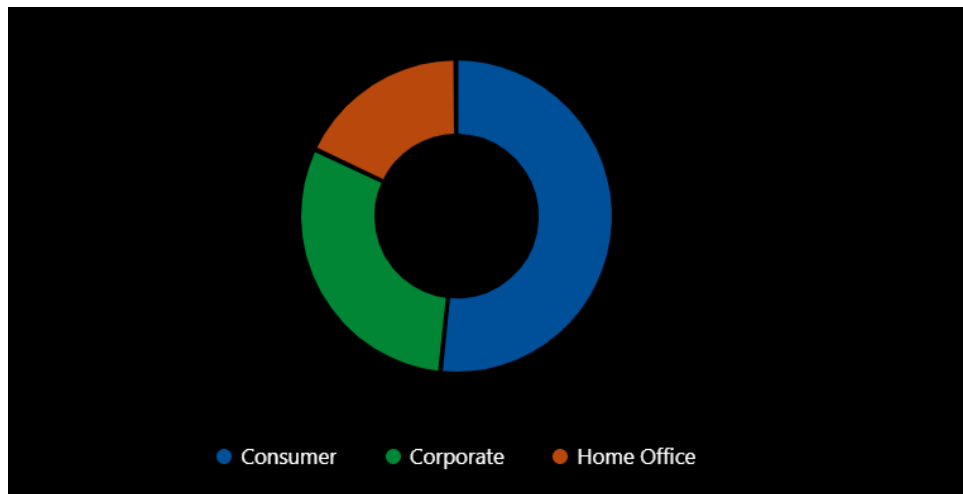


Figure 5.2: Customer Segment Distribution

5.1.3 Shipping Mode Distribution

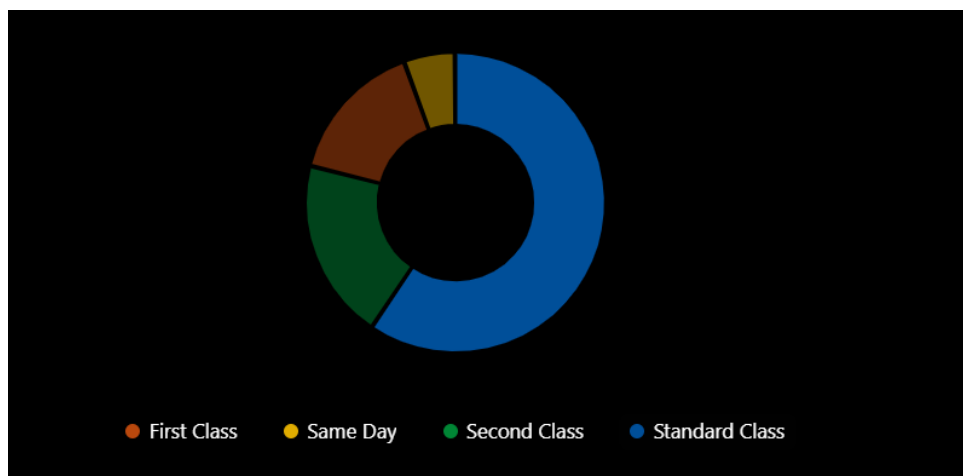


Figure 5.3: Shipping Mode Distribution

5.1.4 Market Distribution

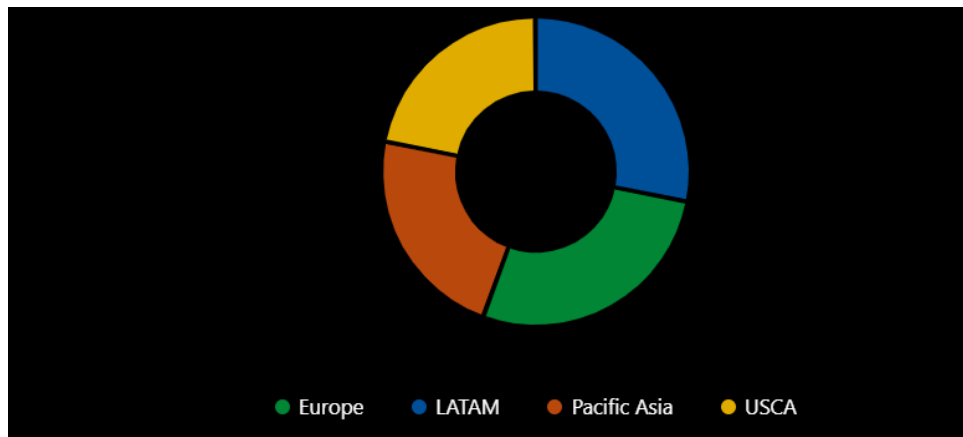


Figure 5.4: Market Distribution

5.1.5 Top Product Categories



Figure 5.5: Top Product Categories

5.2 Data Relationships

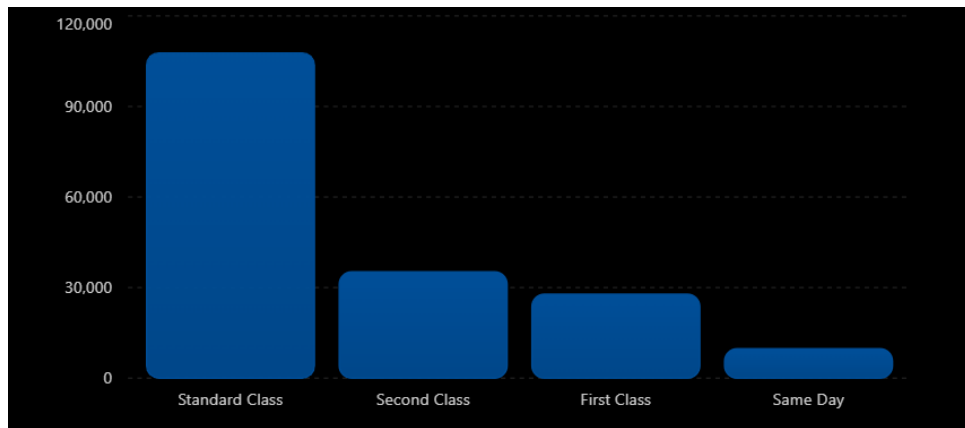


Figure 5.6: Shipping Mode Usage

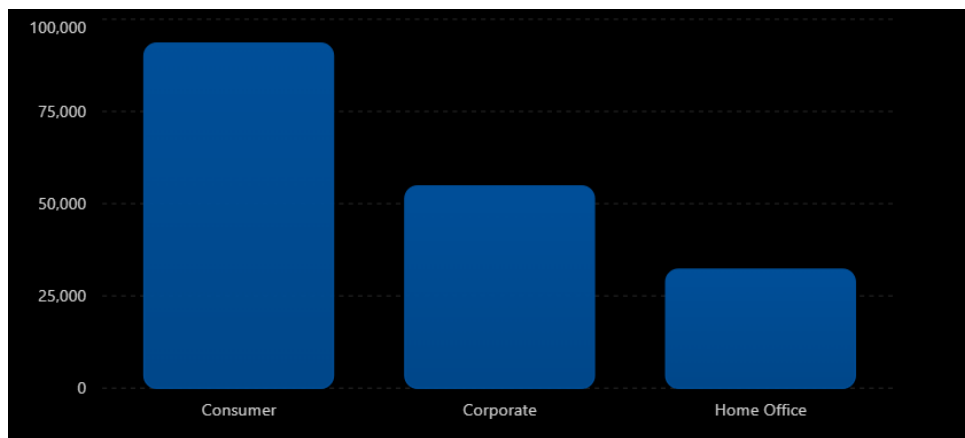


Figure 5.7: Customer Segment Orders

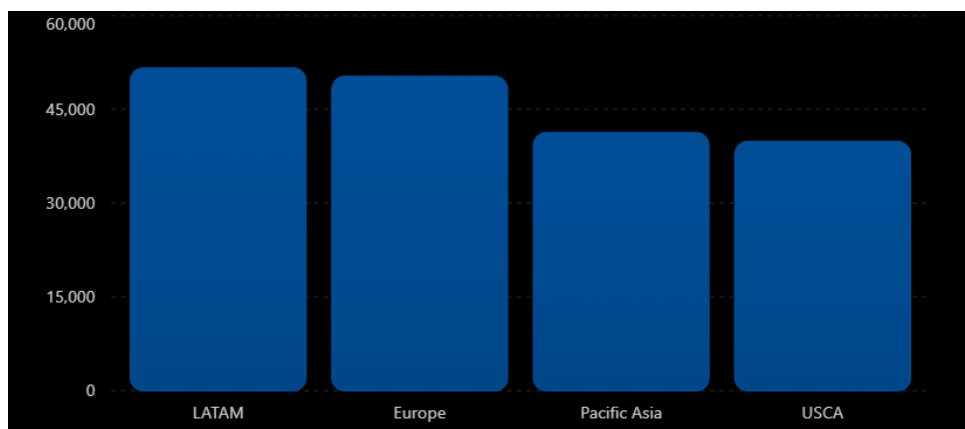


Figure 5.8: Market-wise Orders

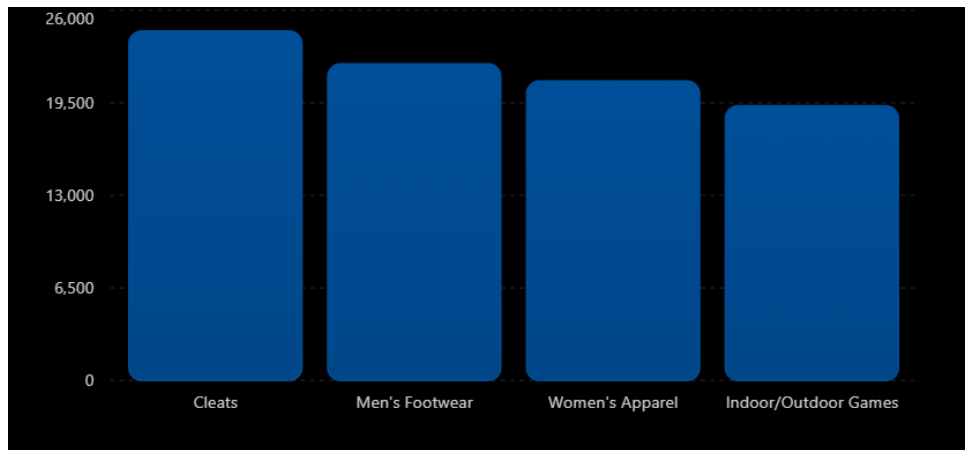


Figure 5.9: Top Product Categories



Figure 5.10: Delivery Status Comparison

5.3 Key Metrics Identification

1. Late deliveries are very high.
2. Standard Class is the most popular shipping mode.
3. Consumer segment dominates total orders.
4. Corporate segment is the second-largest customer group.
5. Home Office segment has the fewest orders.
6. LATAM market contributes the highest order volume.
7. Europe is a major market contributor.
8. Pacific Asia shows strong customer demand.

9. Cleats are the best-selling product category.
10. Men's Footwear has high sales volume.
11. Women's Apparel remains highly popular.
12. Indoor/Outdoor Games generate strong sales.
13. Overall sales exceed \$36 million.
14. Overall profit remains positive.
15. Late delivery risk affects a large number of orders.

Chapter 6

Conclusion

The project successfully analyzed supply chain operations using the DataCo Supply Chain Dataset and generated valuable insights regarding sales performance, customer behavior, shipping operations, and market trends. The analysis highlighted critical business areas requiring improvement and provided recommendations for supply chain optimization.

Chapter 7

Future Scope

- Predict shipment delays using Machine Learning.
- Forecast future sales trends.
- Develop interactive dashboards.
- Optimize inventory management.
- Implement advanced supply chain analytics.
- Enhance demand forecasting capabilities.

References

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3. Pandas Documentation
4. NumPy Documentation
5. Matplotlib Documentation
6. Seaborn Documentation
7. Google Colab Documentation
8. GitHub Documentation

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